FIDEL SOTO GONZÁLEZ

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Summary

Passionate and highly qualified STEM educator and researcher with a PhD in Biomedical Sciences, specializing in microbiology and molecular biology. Experienced in teaching, mentoring, and training diverse student populations, including underrepresented minorities, across academic and research settings. Proven ability to develop innovative curricula, personalize learning experiences, and inspire curiosity in science. Skilled in advanced research techniques such as CRISPR/Cas9, protein analysis, and bioassay development, with a strong publication record in mitochondrial biology and wound healing. Dedicated to fostering global STEM capacity through education, mentorship, and outreach initiatives that bridge gaps in access to scientific knowledge.

This version emphasizes your teaching experience, mentorship of minority students, research expertise, and alignment with Science Corps' mission to build STEM capacity globally. It also highlights your ability to inspire students while leveraging advanced scientific skills.

Relevant Scientific skills and professional skills

- Cell Culture
- Protein Purification
- CRISPR/Cas9
- Project Management & Multitasking
- DNA sequencing for library preparation
- DNA/RNA Extraction
- PCR and RT-PCR
- Exposure Assessment & Risk Evaluation
- Western Blot
- Molecular Cloning
- Effective Presentation
- Mentoring and Coaching
- Bioassay Development
- Mouse Surgery

Education

PhD, Biomedical Sciences (Microbiology), Meharry Medical College (2023)
Bachelor of Science in Biology (Biomedical Science) University of Puerto Rico Aguadilla (2015)

Professional Research Experience

Postdoctoral Fellow *Baylor College of Medicine* (April 2024- Present)

- Led research activities, evaluated and prioritized external research opportunities, and coordinated communication with internal teams of 3 or more scientists.
- Develop protein-protein interactions projects related to male sperm infertility.
- Collaborated with external institutions for research projects, demonstrating strong interpersonal and communication skills.
- Collaborated with data scientists and bioinformaticians to better implement processes related to data analysis for genomic library preparation of patients.

Postdoctoral Fellow *NIH; National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) Institute* (Jan 2023- March 2024)

- Led research activities, evaluated and prioritized external research opportunities, and coordinated communication with internal teams of 3 or more scientists.
- Develop *C. elegans* cell lines using CRISPR Knock-in technology, showcasing expertise in gene editing.
- Collaborated with external institutions for research projects, demonstrating strong interpersonal and communication skills.
- Image *C. elegans* embryos to characterize novel features in cell division.

Graduate Research Assistant *Meharry Medical College* (Aug 2017– Dec 2022)

- Directed PhD thesis characterizing TIM protein complexes in mitochondria biology using biochemical, molecular biology, bioinformatics, and protein analysis techniques, resulting in publication with implications in clinical outcomes for infectious diseases (Human African Trypanosomiasis disease) to potentially target unique protein-protein interactions in parasites while avoiding off-targeting human protein-protein interactions.
- Developed a protein-protein interactions BioID assay, enhancing the ability to detect mitochondria neighboring proteins and identification of novel TIM proteins and monitor Cell line maintenance.
- Assisted in developing Immuno-Assays for NK-Cells and T cells for cytokine release to study cellular immunobiology for immune-based anti-tumor responses using Western blot.
- Mentored 2 PhD students; coached 85 graduate students leading to teaching assistant position.
- Trained 2 medical students in protein analysis techniques resulting in new scientific hypothesis.

Research Assistant *the Ohio State University* (Oct 2015 – Jul 2017)
- Measure mice open wounds and isolated immune cells for primary cell culture to study diabetic conditions under different treatments with drugs, which resulted in 2 publications.
- Maintained, genotyped, monitored mice colonies for surgery preparation, and performed survival surgery on mice while reporting to a matrix oriented group of scientists to report progress in research scientific findings on a fast-paced environment with PCR on a daily basis.
- Trained volunteer students in how to perform genotyping of colonies, mice handling, and normal aseptic laboratory techniques and habits.

Laboratory Technician *Puerto Rico Aqueduct and Sewage Authority* (January 2015 – May 2015)
- Determined water sample quality and made recommendations for potable water expanding my interest in pursuing a PhD in microbiology.
- Collected water samples from different water stations to measure their organic solids content in different checkpoint stations.

Research Assistant *University of Puerto Rico at Aguadilla* (Jan 2014 – May 2014)
- Performed pharmacological studies to understand how different pesticide chemicals affected the lizard *Anolis pulchellus* and asses the effects using Protein Analysis Techniques.
- Studied how estradiol, a component of some pesticides and herbicides, could affect testosterone in male lizards.
- Lizards were used as sentinel organisms to measure how pesticides and herbicides can affect agriculture personnel.

Leadership Experience

Adjunct Professor *Department of Biology, San Jacinto Community College* (Jan 2025 – Present)
- Instructed laboratory and classes in Biology, Anatomy and Physiology for undergraduate students.
- Met with professors to address concerns and complicated topics covered in class that required more attention to detail.
- Designed and implemented strategies to optimize group learning in different class settings.

Academic Tutor *Center for Educational Development and Support, Meharry Medical College* (Oct 2018 – Dec2022)
- Consultant, small group session facilitator, and instructor leading improvement in areas of Immunobiology, Immunometabolism, and Microbiology on cancer stem cells, aging.
- Met with professors to address concerns and complicated topics covered in class that required more attention to detail.
- Designed and implemented strategies to optimize group learning in different class settings.

Teaching Assistant *Meharry Medical College* (Spring 2020)
- Designed and directed journal club styled classes to discuss scientific findings and how they could be applied with laboratory techniques in terms of Cell Biology and Immunobiology cancer stem cells, aging.
- Evaluated and studied students’ progress during the courses to understand milestones and improvements needed to update the course dynamics and style.

Awards

First Year PhD Award (Best Student of the Year) *Meharry Medical College* (2018)
The Reuben and Hermine Herzfeld Scholarship (Outstanding GPA) *Meharry Medical College* (2018)
The Charles W. Johnson Sr. M.D. Endowed Scholarship (Outstanding Research) *Meharry Medical College* (2019)
Martin Luther King Jr. (Outstanding Community Contributions) *Meharry Medical College* (2021)
Postdoctoral Fellowship *National Institute of Diabetes and Digestive and Kidney Diseases (2023)*

Publications

- Quiñones Guillén, Linda S., Fidel Soto Gonzalez, Chauncey Darden, Muhammad Khan, Anuj Tripathi, Joseph T. Smith, Ayorinde Cooley et al. "Unique Interactions of the Small Translocases of

the Mitochondrial Inner Membrane (Tims) in Trypanosoma Brucei." International Journal of Molecular Sciences 25, no. 3 (2023). 10.3390/ijms25031415.

- Soto-Gonzalez, F., A. Tripathi, A. Cooley, V. Paromov, T. Rana, and M. Chaudhuri. "A Novel Connection between Trypanosoma Brucei Mitochondrial Proteins Tbtim17 and Tbtrap1 Is Discovered Using Biotinylation Identification (Bioid)." J Biol Chem 298, no. 12 (Oct 26 2022): 102647.
- Chaudhuri, M., A. Tripathi, and F. S. Gonzalez. "Diverse Functions of Tim50, a Component of the Mitochondrial Inner Membrane Protein Translocase." Int J Mol Sci 22, no. 15 (Jul 21 2021). http://dx.doi.org/10.3390/ijms22157779.
- Chaudhuri, M., C. Darden, F. S. Gonzalez, U. K. Singha, L. Quinones, and A. Tripathi. "Tim17 Updates: A Comprehensive Review of an Ancient Mitochondrial Protein Translocator." Biomolecules 10, no. 12 (Dec 7 2020). http://dx.doi.org/10.3390/biom10121643.
- Gnyawali, S. C., M. Sinha, M. S. El Masry, B. Wulff, S. Ghatak, F. Soto-Gonzalez, T. A. Wilgus, S. Roy, and C. K. Sen. "High Resolution Ultrasound Imaging for Repeated Measure of Wound Tissue Morphometry, Biomechanics and Hemodynamics under Fetal, Adult and Diabetic Conditions." PLoS One 15, no. 11 (2020): e0241831. http://dx.doi.org/10.1371/journal.pone.0241831.
- Singh, K., M. Sinha, D. Pal, S. Tabasum, S. C. Gnyawali, D. Khona, S. Sarkar, S. K. Mohanty, F. Soto-Gonzalez, S. Khanna, S. Roy, and C. K. Sen. "Cutaneous Epithelial to Mesenchymal Transition Activator Zeb1 Regulates Wound Angiogenesis and Closure in a Glycemic StatusDependent Manner." Diabetes (Aug 22 2019). http://dx.doi.org/10.2337/db19-0202.

Oral Presentations

- *Hey Tim, This a TRAP! How BioID found TbTim17 neighboring TbTRAP1.* October 2022. Cellular Biology of Eukaryotic Pathogens Conference. Clemson University, South Carolina
- *BioID Approach Revealed TbTRAP-1 as a Proximal Partner of TbTim17*. 2021. 63rd Annual Student Research Day. Nashville, TN.
- *Characterization of the Composition and Arrangement of the Components of the TIM Complex in Trypanosoma brucei.* 2019, April 23. West Basic Science Building, Meharry Medical College. Nashville, Tennessee
- *Neutrophils Enhance Early Trypanosoma brucei Infection Onset.* 2018, *August 21.* West Basic Science Building, Meharry Medical College. Nashville, Tennessee